



REGULATIONS 2024
FLEXIBLE CHOICE BASED CREDIT SYSTEM
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
B.E. Electronics and Communication Engineering
CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Exempted for LE	HS	SH1106	19EN106	English for Engineers	-	3	1	4	CGPA	TIP (6 Expts)
2		HS	SH1107	19EN107	Technical Writing and Presentation	-	0	2	2	CGPA	Practical (10 Expts)
3	Exempted for LE	HS	SH9801	NA	Organizational Behavior and Leadership	-	3	0	3	CGPA	Theory
4		HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
5	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
6	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
7	Exempted for LE	BS	SH3216	19PH216	Physics for Electronics Engineering	-	3	1	4	CGPA	TIP (6 Expts)
8	Exempted for LE	BS	SH4805	NA	Chemistry of Engineering Materials	-	2	1	3	CGPA	TIP (6 Expts)
9	Exempted for LE	BS	SH2201	19MA201	Calculus and Matrix Algebra	-	4	0	4	CGPA	Theory
10		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
11	Exempted for LE	BS	SH2223	19MA223	Linear Algebra and Probability	-	4	0	4	CGPA	Theory
12	Exempted for LE	BS	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
13	Exempted for LE	ES	EC2301	19EE301	Electric Circuits and Networks	19MA202	3	1	4	CGPA	TIP (6 Expts)
14		ES	CS3304	19AI304	Fundamentals of C Programming	-	2	1	3	CGPA	TIP (6 Expts)
15		ES	CS3305	19AI305	Advanced C Programming	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
16	Exempted for LE	ES	CS3408	19AI408	Data Structures	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
17	Any 1	ES	CS3306	19AI306	Object Oriented Programming using C++	-	2	1	3	CGPA	TIP (6 Expts)
18	Any 1	ES	CS3307	19AI307	Object Oriented Programming using Java	19AI304	2	1	3	CGPA	TIP (6 Expts)
19		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
20		ES	EC1302	19EC302	Electronic Devices	-	3	1	4	CGPA	TIP (6 Expts)
21		PC	EC1402	19EC402	Signals and Systems	19MA202	3	0	3	CGPA	Theory
22		PC	EC1801	NA	Digital Logic Circuits Design	-	3	1	4	CGPA	TIP (6 Expts)

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23		PC	EC1419	19EC419	Electronic Circuits	-	3	1	4	CGPA	TIP (6 Expts)
24		PC	EC1802	NA	Electromagnetic Theory and Transmission Lines	-	4	0	4	CGPA	Theory
25		PC	EC1421	19EC421	Analysis and Design of Analog ICs	19EC302 / 19EE405 / EC1302 / 19EC308	3	1	4	CGPA	TIP (6 Expts)
26		PC	EC1408	19EC408	Microprocessor and Microcontroller	19EC303 / EC1801	3	1	4	CGPA	TIP (6 Expts)
27		PC	EC1420	19EC420	Analog Communication	19EC402 / EC1402	3	1	4	CGPA	TIP (6 Expts)
28		PC	EC1406	19EC406	Digital Communication	19EC403 / EC1420	3	1	4	CGPA	TIP (6 Expts)
29		PC	EC1409	19EC409	Discrete Time Signal Processing	19EC402 / EC1402	3	1	4	CGPA	TIP (6 Expts)
30		PC	EC1410	19EC410	VLSI Design	19EC303 / EC1801 / EC2404	3	1	4	CGPA	TIP (6 Expts)
31		PC	EC1413	19EC413	Wireless Communication	19EC406 / EC1406	3	0	3	CGPA	Theory
32		PC	EC1424	19EC424	Embedded Systems and IoT	19EC408 / EC1408	2	1	3	CGPA	TIP (6 Expts)
33		PC	EC1823	19EE312	Control System Engineering	19EC402 / EC1402	3	0	3	CGPA	Theory
34		PC	EC1425	NA	Optical Communication and Network	EC1406	2	1	3	CGPA	TIP (6 Expts)
35		PC	EC1418	NA	Communication Networks and Cyber Security	EC1406	2	1	3	CGPA	TIP (6 Expts)
36		PC	EC1411	19EC411	Antennas and Microwave Engineering	19EC407	3	1	4	CGPA	TIP (6 Expts)
37		PC	EC1520	19EC520	Introduction to MEMS and NEMS	-	3	0	3	CGPA	Theory
38		PE(Honours)	EC1523	19EC523	Verilog HDL	19EC303 / EC1801 / EC2404	2	1	3	CGPA	TIP (6 Expts)
39		PE(Honours)	EC1521	19EC521	System Verilog for Design and Verification	19EC523 / EC1523	2	1	3	CGPA	TIP (6 Expts)
40		PE(Honours)	EC1522	19EC522	Physical Design for VLSI	19EC410 / EC1410	2	1	3	CGPA	TIP (6 Expts)
41		PE(Honours)	EC1822	19EC822	VLSI Technology	EC1410 / 19EC410	3	0	3	CGPA	Theory
42		PE(Honours)	EC1517	19EC517	Low Power VLSI Design	19EC410 / EC1410	3	0	3	CGPA	Theory
43		PE(Honours)	EC1519	19EC519	Testing of VLSI Circuits	19EC410 / EC1410	3	0	3	CGPA	Theory
44		PE(Honours 1)	EC1524	19EC524	Natural Language Processing	19EC409 / EC1409	3	0	3	CGPA	Theory
45		PE(Honours 1)	EC1503	19EC503	Digital Image Processing	19EC409 / EC1409	3	0	3	CGPA	Theory
46		PE(Honours 1)	EC1803	NA	Computer Vision and Pattern Recognition	-	3	0	3	CGPA	Theory
47		PE(Honours 1)	CS1557	19CS557	Software Defined Networks	-	2	1	3	CGPA	TIP (6 Expts)
48		PE(Honours 1)	EC1804	NA	Wireless Sensor Network Design	EC1421	3	0	3	CGPA	Theory

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49		PE(Honours 1)	EC1805	NA	Data Storage Technologies	-	3	0	3	CGPA	Theory
50		PE(Honours 1)	EC1504	19EC504	DSP Processors	19EC409	3	0	3	CGPA	Theory
51		PE(Honours 2)	EC1806	NA	Software for Embedded System	-	3	0	3	CGPA	Theory
52		PE(Honours 2)	EC1807	NA	Real Time Operating Systems	-	3	0	3	CGPA	Theory
53		PE(Honours 2)	EC1808	NA	Automotive Embedded Systems	EC1424	3	0	3	CGPA	Theory
54		PE(Honours 2)	EC1809	NA	IoT Processors	EC1424	2	1	3	CGPA	TIP (6 Expts)
55		PE(Honours 2)	EC1810	NA	IoT Based System Design	EC1424	3	0	3	CGPA	Theory
56		PE(Honours 3)	EC1811	NA	5G Communication and Networks	-	2	1	3	CGPA	TIP (6 Expts)
57		PE(Honours 3)	EC1812	NA	5G Edge and Cloud Computing	-	2	1	3	CGPA	TIP (6 Expts)
58		PE(Honours 3)	EC1813	NA	5G Blockchains and Security	-	3	0	3	CGPA	Theory
59		PE(Honours 3)	EC1814	NA	Smart Healthcare Systems	-	3	0	3	CGPA	Theory
60		PE(Honours 3)	EC1815	NA	AR and VR for 5G	-	2	1	3	CGPA	TIP (6 Expts)
61		PE(Honours 3)	EC1816	NA	6G Communication Technologies	-	3	0	3	CGPA	Theory
62		PE(Honours 3)	EC1511	19EC511	Satellite Communication	19EC406 / EC1406	3	0	3	CGPA	Theory
63		PE(Honours 4)	EC1817	NA	MEMS Fabrication and Integration	-	2	1	3	CGPA	TIP (6 Expts)
64		PE(Honours 4)	EC1818	NA	FEA and MEMS Simulation	-	2	1	3	CGPA	TIP (6 Expts)
65		PE(Honours 4)	EC1525	19EC525	RF MEMS and MOEMS	19EC520	2	1	3	CGPA	TIP (6 Expts)
66		PE(Honours 4)	EC1819	NA	BioMEMS and Microfluidics	-	2	1	3	CGPA	TIP (6 Expts)
67		PE(Honours 4)	EC1820	NA	Nanoscience and Nano Technology	-	2	1	3	CGPA	TIP (6 Expts)
68		PE(Honours 4)	EC1821	NA	Design, Testability and Applications of MEMS and NEMS	-	3	0	3	CGPA	Theory
69		PE	CS3404	19AI404	Analysis of Algorithms	19AI301	2	1	3	CGPA	TIP (6 Expts)
70		PE	CS1404	19CS404	Database Management System and its Applications	19AI301/19AI304/19CS301/19CS302/19CS303	3	1	4	CGPA	TIP (6 Expts)
71		PE	CS1509	19CS509	Graph Theory and Applications	19MA206	3	0	3	CGPA	Theory
72		PE	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
73		PE	CS3801	NA	Artificial Intelligence and Applications	-	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)

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74		PE	CS1542	19CS542	Embedded Board Design	-	2	1	3	CGPA	TIP (6 Expts)
75		PE	CS1543	19CS543	Embedded Board Manufacturing	-	2	1	3	CGPA	TIP (6 Expts)
76		EEC	EC1701	19EC701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
77	Option 1	EEC	EC1702	19EC702	Project Work Phase I	-	0	2	2	CGPA	Project (Int Only)
78	Option 1	EEC	EC1703	19EC703	Project Work Phase II	19EC702	0	6	6	CGPA	Project
79	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
80		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
81		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
82		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
83		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
84		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
85		EEC	SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711/SH6711	1	0	1	CGPA	Theory (Int Only)
86		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
87		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
88		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
89	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
90	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
91	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
92	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
93	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	13	4
Basic Science	21	4
Engineering Science	23	16
Professional Core	61	61
Professional Electives	15	15
Open Electives	12	12
Employability Enhancement Courses	16	16
Mandatory Courses	3	3
TOTAL	164	131

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